

**SPECIAL TECHNICAL REPORT |  
SEISMIC ENGINEERING  
STRUCTURAL PERFORMANCE ·  
DISASTER RESILIENCE · LESSONS FOR INDIA**

**LESSONS FROM THE  
2026 MINDANAO  
EARTHQUAKE**

Engineering Insights, Structural Performance & Future Resilience

**Mw 7.8**  
Magnitude

**Sarangani,  
Mindanao**  
Location

**08 June 2026**  
Date of Event

**47 Confirmed  
Deaths**  
As of 11 Jun 2026

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*Prepared by*  
**Hiren J Shah**  
**Structural Engineer**  
FormX Consultants LLP  
Design | Engineering

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## 1. Executive Summary

At 07:37 AM Philippine Standard Time on Monday, 08 June 2026 — the first day of the new school year for more than three million children across Mindanao — a moment magnitude Mw 7.8 earthquake struck offshore Sarangani Province in the southern Philippines. The earthquake's epicentre was located approximately 32 kilometres west-southwest of Maasim, Sarangani, at a verified depth of 33 kilometres (PHIVOLCS). It was caused by subduction along the Cotabato Trench — the same seismic structure that produced the catastrophic Moro Gulf earthquake of 17 August 1976 (Mw 8.0), which killed approximately 8,000 people.

The 2026 event is the most powerful earthquake to strike the Philippines since the July 1990 Luzon earthquake (also Mw 7.8), which killed more than 1,600 people and destroyed much of Baguio City. The 2026 event confirmed dead at **47 persons, 688 injured, and 33 missing as of 11 June 2026**, according to the National Disaster Risk Reduction and Management Council (NDRRMC). Across Mindanao, **more than 12,600 houses were damaged or destroyed**. Eighteen bridges in Soccsksargen were damaged by DPWH reporting, with six rendered near-impassable. **At least 41 roads were affected, approximately 20 rendered impassable**. The international airport in General Santos City was shut for at least two days, cancelling 63 domestic flights.

The earthquake triggered a tsunami warning issued for nine provinces across the southern Philippines — Sarangani, Davao Occidental, Tawi-tawi, Sulu, Basilan, Zamboanga del Sur, Zamboanga Sibugay, Sultan Kudarat, and South Cotabato — along with international warnings for Indonesia, Malaysia, Palau, Japan, Taiwan, and Papua New Guinea. Tsunami wave heights of approximately 1 metre were recorded along the coasts of Kiamba and Maasim in Sarangani, and Kalamansig in Sultan Kudarat. Smaller waves were measured in Indonesia, Palau, and southern Japan. The Pacific Tsunami Warning Center lifted the tsunami threat approximately five hours after the mainshock.

Approximately 917,000 people were within the zone of MMI VIII (Severe) shaking, and 2.66 million more were exposed to MMI VII (Very Strong) shaking across Soccsksargen, Davao Region, and Bangsamoro (USGS PAGER). This is a vastly larger exposed population than the 2025 Cebu event, underscoring the 2026 event's significance as a major national disaster.

| Parameter             | Verified Data                                                                            |
|-----------------------|------------------------------------------------------------------------------------------|
| Event Date & Time     | 08 June 2026, 07:37:43 AM PST (23:37:43 UTC, 07 June)                                    |
| Magnitude             | Mw 7.8 — PHIVOLCS (upgraded from initial Mw 7.0), USGS & EMSC; Mw 7.7 per BMKG Indonesia |
| Epicentre             | 32 km west-southwest of Maasim, Sarangani Province (05.57°N, 124.98°E)                   |
| Depth                 | 33 km (PHIVOLCS); 55.2 km (USGS); 45 km (EMSC); 47 km (BMKG)                             |
| Causative Structure   | Subduction along the Cotabato Trench (PHIVOLCS confirmed)                                |
| Mechanism             | Subduction / oblique convergence — Cotabato Trench                                       |
| Max. Intensity (PEIS) | PEIS VII (Destructive) — General Santos City                                             |

|                           |                                                                                                                   |
|---------------------------|-------------------------------------------------------------------------------------------------------------------|
| Max. Intensity (MMI)      | MMI VIII (Severe) — estimated by USGS PAGER                                                                       |
| Population Exposed        | ~917,000 in MMI VIII zone; ~2.66 million in MMI VII zone                                                          |
| Aftershocks (by 11:00 AM) | 138 recorded M1.3–M6.7 (PHIVOLCS); strongest M6.7                                                                 |
| Significant Aftershocks   | M6.5, M6.0, M6.0 (USGS) — within hours of mainshock                                                               |
| Tsunami Height Recorded   | ~1.0–1.4 m — Kiamba & Maasim (Sarangani), Sultan Kudarat                                                          |
| Tsunami Warnings Issued   | 9 Philippine provinces; Indonesia, Malaysia, Palau, Japan, Taiwan, PNG                                            |
| Confirmed Deaths          | 47 (NDRRMC, as of 11 June 2026)                                                                                   |
| Breakdown                 | 19 — Sarangani; 15 — South Cotabato (incl. 13 in GenSan); 11 — Davao Occidental; 1 — Davao del Sur; 3 — Indonesia |
| Confirmed Injuries        | 688 (NDRRMC); 129 in Soccsksargen; 5 in Davao Region (early figures)                                              |
| Missing Persons           | 33 — 18 Davao Occidental; 12 Sarangani; 4 South Cotabato                                                          |
| Houses Damaged/Destroyed  | More than 12,600 across Mindanao                                                                                  |
| Bridges Damaged           | 18 in Soccsksargen (6 near-impassable) — per DPWH initial report                                                  |
| Roads Affected            | 41 (approx. 20 impassable) — per DPWH initial report                                                              |
| Landslides                | At least 6 confirmed (Glan, Jose Abad Santos, Davao del Sur sites)                                                |
| Airport Impact            | General Santos International Airport closed ≥2 days; 63 domestic flights cancelled                                |
| Schools for Assessment    | ~6,000 public school buildings in quake-hit provinces                                                             |

**Table 1.1 — Verified Event Parameters & Impact Data (Sources: PHIVOLCS/DOST Primer 08 Jun 2026; USGS/ANSS; NDRRMC; OCHA; DPWH; GMA News; Philippine Inquirer)**

### ENGINEERING INSIGHT — This is the Most Powerful Philippine Earthquake Since 1990

The Mw 7.8 Sarangani earthquake is the strongest to strike the Philippines since the 16 July 1990 Luzon earthquake (Mw 7.8), which killed 1,621 people. It is also the largest event generated by the Cotabato Trench since at least 1976. The 1976 Moro Gulf earthquake (Mw 8.0–8.1) on the same tectonic structure killed approximately 8,000 people. The 2026 event must be understood within this seismic history — the Cotabato Trench is one of the Philippines' most formidable seismic hazard sources, yet it has received less engineering attention than the Philippine Fault Zone and Manila Trench.

## 2. Understanding the Earthquake

### 2.1 Regional Tectonic Setting

The southern Philippines sits within one of the most tectonically complex regions on Earth — the convergence zone between the Philippine Sea Plate, the Eurasian Plate, and the Indo-Australian Plate. Mindanao, the Philippines' second-largest island, is bounded by several major subduction zones and associated fault systems that collectively generate a large fraction of global seismic energy.

The primary tectonic structures governing seismicity in southern Mindanao include: the Philippine Trench to the east (a major west-dipping subduction zone); the Cotabato Trench to the southwest (where the Australian Plate subducts beneath the Philippine micro-plate); and the Cotabato Fault System — a network of NW–SE trending sinistral (left-lateral) strike-slip faults that form the structural boundary between the Cotabato Arc and the Central Mindanao volcanic belt.

### 2.2 The Cotabato Trench — Causative Structure

PHIVOLCS confirmed in its official primer that the 2026 earthquake 'may be attributed to subduction along the Cotabato Trench,' based on the event's location, focal mechanism, and aftershock distribution. The Cotabato Trench is a major earthquake-generating structure that accommodates part of the oblique convergence between the Australian Plate and the Philippine micro-plate. Subduction along this trench generates thrust-type earthquakes with significant tsunami-generating potential.

The 1976 Moro Gulf earthquake — the most destructive natural disaster in Philippine recorded history, killing approximately 8,000 people — was generated by the same tectonic system. The 2026 event, while significantly smaller, demonstrates that this system remains seismically active and is capable of producing major events with little warning.

### 2.3 Why This Earthquake Generated a Tsunami

Subduction zone earthquakes generate tsunamis when vertical seafloor displacement occurs during fault rupture. Unlike the 2025 Cebu earthquake (strike-slip; minimal tsunami), the Cotabato Trench rupture involves thrust faulting — the seafloor is displaced upward by several metres over the rupture area, displacing the overlying water column and generating a series of tsunami waves. The tsunami wave heights of approximately 1 metre recorded at nearby coastlines are consistent with an Mw 7.8 thrust event at 33 km depth and approximately 20 km from shore — not devastating but capable of damaging coastal structures and posing a drowning risk.

### 2.4 Explanation for Non-Technical Readers

Think of the Earth's crust beneath the ocean as two giant slabs, one slowly sliding under the other — like a slow-motion collision between two enormous stone slabs. For decades, stress builds up at the locked interface. When the stress exceeds the friction, the upper slab snaps upward — suddenly and violently. That upward movement lifts billions of tonnes of seawater instantaneously, sending waves outward in all directions. On land, the same energy radiates outward as seismic waves — the violent shaking that damages buildings and triggers landslides. On 8 June 2026, that snap occurred under the sea floor near Sarangani at 7:37 in the morning, while millions of children were starting their first school day.

### FIGURE SUGGESTION — Figure 2.1

Tectonic map of southern Philippines and northern Indonesia showing: Cotabato Trench, Philippine Trench, Philippine Fault Zone, Cotabato Fault System, and the 2026 event star marker (05.57°N, 124.98°E). Overlay with MMI VIII and VII shaking zones from USGS PAGER. Show the 1976 Moro Gulf earthquake epicentre for comparison. Source: PHIVOLCS / USGS.

## 3. Earthquake Timeline

| Date / Time (PST)        | Event                                                                                                                   | Engineering / Response Significance                                                                 |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| 08 Jun 2026,<br>07:37 AM | Mainshock — Mw 7.8, offshore Sarangani Initially reported Mw 7.0 by PHIVOLCS; upgraded to 7.8                           | Onset of structural damage; peak ground motion experienced across Soccsksargen and Davao regions    |
| 07:37–08:00 AM           | PHIVOLCS issues Tsunami Warning for 9 Mindanao provinces; international warnings follow                                 | Coastal evacuation triggered; PCG suspends all watercraft operations                                |
| 07:37–08:30 AM           | First school day of SY 2026–2027; flag ceremonies underway across Mindanao Teachers direct duck-cover-hold protocol     | No casualties in schools executing correct protocol; highlights value of earthquake drills          |
| 08:00–09:00 AM           | 138 aftershocks recorded M1.3–M6.7 by PHIVOLCS; significant M6.5, M6.0, M6.0 aftershocks (USGS)                         | Cascading damage risk to weakened structures; rescue operations complicated by aftershock sequences |
| 08:00 AM onwards         | Glan landslide buries houses — 14 confirmed dead (Sarangani)                                                            | Earthquake-triggered landslide: primary secondary hazard in steep terrain                           |
| 08:00 AM onwards         | Jose Abad Santos landslides — 7 dead, 18 missing; town partially cut off                                                | Access routes blocked; search and rescue severely hindered                                          |
| 08:00 AM onwards         | St. Elizabeth Hospital, General Santos — partial structural damage; patients evacuated; operations relocated outdoors   | Hospital functionality failure: critical lifeline structure compromise during disaster response     |
| 08:00 AM onwards         | Notre Dame of Dadiangas University building collapses partially; Matanao National High School (pre-condemned) collapses | Pre-condemned school collapses: post-earthquake accountability for undemolished unsafe structures   |

|                   |                                                                                                                                                               |                                                                                                       |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| ~08:30 AM         | Tsunami waves of ~1 m recorded at Kiamba, Maasim (Sarangani) and Kalamansig (Sultan Kudarat); waves felt in Indonesia, Palau, Japan                           | Tsunami impact on coastal infrastructure confirmed; six stilted shanties damaged in Zamboanga del Sur |
| Morning–Afternoon | AFP 10th Infantry Division, PNP, Coast Guard, BFP deploy HADR teams; PAF deploys aircraft from Davao                                                          | Multi-agency rapid assessment activated; air transport for officials and engineers                    |
| ~12:30 PM         | Pacific Tsunami Warning Center: destructive tsunami threat largely passed ~5 hours after mainshock                                                            | Staged lifting of coastal evacuation zones                                                            |
| 08 Jun, PM        | GenSan International Airport closed; 63 domestic flights cancelled                                                                                            | Supply chain and rescue logistics disruption                                                          |
| 09 Jun 2026       | President Marcos visits General Santos; distributes financial assistance; orders all agencies to act immediately                                              | National government mobilisation confirmed; QRF funds activated                                       |
| 09–10 Jun 2026    | DepEd: ~6,000 school buildings in quake-hit provinces require assessment before classes resume DPWH: 18 bridges damaged, 6 near-impassable; 41 roads affected | Scale of infrastructure damage assessment defined                                                     |
| 10 Jun 2026       | VP Sara Duterte attends victims' wake in Malapatan, Sarangani National Historical Commission issues heritage site safety reminders                            | Heritage structure risk advisory: first formal post-event heritage alert                              |
| 11 Jun 2026       | NDRRMC confirms: 47 dead, 688 injured, 33 missing 12,600+ houses damaged / destroyed                                                                          | Casualty figures stabilising; damage assessments ongoing                                              |

**Table 3.1 — Earthquake and Emergency Response Timeline (Sources: PHIVOLCS Primer 08 Jun 2026; NDRRMC SitReps; OCHA UN News; GMA News; Philippine Inquirer; NPR; Al Jazeera)**

## 4. Ground Motion and Seismic Characteristics

### 4.1 Intensity Distribution

PHIVOLCS confirmed a maximum intensity of PEIS VII (Destructive) in General Santos City using instrumental records (Earthquake Information No. 2 of 08 June 2026). Shaking was felt as far as Abuyog

and Dulag in Leyte Province at PEIS II (Slightly Felt) — a propagation distance exceeding 700 km, confirming the event's exceptional magnitude. Instrumental intensity estimates from the USGS ShakeMap placed MMI VIII (Severe) across a zone encompassing approximately 917,000 people, with MMI VII (Very Strong) affecting a further 2.66 million across Soccsksargen, Davao, and Bangsamoro regions.

The relatively moderate PEIS VII at General Santos City — a city of approximately 722,000 people located some 15–20 km from the epicentre — compared to the broader MMI VIII estimate is consistent with a deeper focal depth (33–55 km) compared to the shallow 2025 Cebu event (5–10 km). Greater focal depth means more energy has dissipated geometrically by the time ground motion reaches the surface, reducing peak amplitudes but broadening the affected area.

## 4.2 Peak Ground Acceleration and Focal Depth Context

Confirmed strong-motion records for this event are being processed by PHIVOLCS and USGS. The focal depth reported by different agencies ranges from 33 km (PHIVOLCS) to 55 km (USGS). This depth range — intermediate-depth subduction zone event — is significantly deeper than the 2025 Cebu earthquake (5–10 km). This difference has a direct and important engineering implication: at greater depth, high-frequency ground motions attenuate more rapidly, while lower-frequency energy propagates to greater distances. This means: (a) less localised extreme shaking close to the epicentre compared to a shallow event; (b) more widespread moderate shaking affecting a much larger area; and (c) lower PGA values near the epicentre than might be expected from a shallow Mw 7.8.

Estimated PGA values at General Santos City, consistent with PEIS VII ground motion and the observed building damage distribution, are in the range of 0.2–0.4 g, which — for inadequately designed or maintained structures — is sufficient to cause collapse.

## 4.3 Tsunami Hazard: Wave Generation, Height, and Impact

The Cotabato Trench thrust mechanism produced vertical seafloor displacement that generated a regional tsunami. Wave heights of approximately 1.0 to 1.4 metres above normal tide level were recorded along the Sarangani and Sultan Kudarat coastlines by PHIVOLCS. These heights, while below catastrophic levels, were sufficient to damage coastal structures, shanties on stilts, and port infrastructure. PHIVOLCS had issued a Destructive Tsunami Warning for nine provinces; the warning was ultimately consistent with the observed wave heights, which were significant but not catastrophic at these distances.

Smaller waves were recorded in Indonesia's North Sulawesi (Sangihe and Talaud Islands), Palau, and southern Japan. The multi-country tsunami alert — involving warnings from PHIVOLCS, BMKG (Indonesia), the Pacific Tsunami Warning Center, and the US National Weather Service — demonstrated effective international tsunami warning coordination, though the 30-minute warning window between earthquake and first wave arrival at the nearest coastlines underscores the irreducible constraint of physics on tsunami evacuation lead times.

## 4.4 Landslide Hazard

At least six landslides were triggered by the earthquake, concentrated in the steep terrain of Glan and Jose Abad Santos in Sarangani Province, and in Kiblawan and Matanao in Davao del Sur. The Glan landslide alone killed 14 people; the Jose Abad Santos landslides killed 7 and left 18 missing, with some



areas of the town cut off from road access as of 10 June. This pattern — where earthquake-induced landslides account for a disproportionate share of casualties compared to direct structural collapse — is consistent with findings from major subduction zone earthquakes in mountainous terrain globally.

#### **FIGURE SUGGESTION — Figure 4.1**

PHIVOLCS ShakeMap showing PEIS isoseismal contours (PEIS II through VII) with the 2026 epicentre marker. Overlay USGS PAGER MMI VIII and VII zones. Mark locations of landslides (Glan, Jose Abad Santos, Davao del Sur), tsunami-affected coastlines (Kiamba, Maasim, Kalamansig), and key damaged structures. Source: PHIVOLCS / USGS PAGER.

## **5. Structural Performance Assessment**

### **5.1 Assessment Programme**

Structural assessment following the 2026 Mindanao earthquake was conducted by multiple agencies simultaneously. The Department of Public Works and Highways (DPWH) prioritised bridges, roads, hospitals, and schools; DPWH Secretary Vince Dizon confirmed immediate deployment of engineering teams for rapid assessment of critical infrastructure. The Department of Education deployed DepEd engineers to assess approximately 6,000 school buildings across quake-hit provinces before authorising resumption of classes. The Philippine Air Force deployed rotary and fixed-wing aircraft from Tactical Operations Group 11 in Davao City for aerial damage assessment and logistics support. The Philippine Red Cross was placed on highest alert with structural assessment teams deployed across affected provinces.

### **5.2 Notable Structural Failures**

#### **St. Elizabeth Hospital, General Santos City**

Parts of the St. Elizabeth Hospital sustained severe damage, forcing evacuation of patients and medical personnel. The facility was compelled to temporarily relocate operations outside the main building — a critical failure of a Lifeline Structure that is specifically required by code to remain functional after a design-level seismic event. This failure mirrors the structural performance of hospitals documented in the 2025 Cebu earthquake and is among the most serious engineering failures of the 2026 event. A hospital that cannot function during the disaster response it exists to support represents a compounding catastrophe.

#### **Notre Dame of Dadiangas University, General Santos City**

A building at Notre Dame of Dadiangas University was observed to crumble during the earthquake. Eyewitness accounts — including from university president Br. Manuel de Leon — and photographic evidence confirmed partial structural collapse of at least one building. The university, founded in 1950, has a mixed building inventory of different construction vintages.

#### **Matanao National High School, Davao del Sur**



The previously condemned school building at Matanao National High School collapsed. Critically, this building had already been structurally condemned following damage in the 2019 Cotabato earthquake series — meaning its collapse was entirely foreseeable and preventable. DepEd Davao del Sur confirmed the structure had been formally condemned, raising serious questions about why an occupied school campus contained an undemolished condemned structural building. No casualties were reported at this specific structure.

### **Warehouse Wall Collapse, Century Pacific Food, General Santos City**

Two workers were killed when a warehouse wall at Century Pacific Food's facility collapsed in General Santos. Industrial and warehouse structures — often built with unreinforced or minimally reinforced masonry perimeter walls — are a consistently vulnerable class of structure in Philippine earthquakes. Parapet and perimeter wall collapse has been a documented hazard in every major Philippine earthquake event.

### **Barangay Hall Collapse, Jose Abad Santos, Davao Occidental**

The Barangay Hall in Malalan, Jose Abad Santos collapsed. Government buildings — particularly at the barangay (village) level — are among the most structurally vulnerable in the Philippines' public building inventory, often constructed as pre-code or minimally supervised small RC or masonry structures.

### **Bridge in Tapon, Glan, Sarangani**

A bridge in Tapon, Glan, was destroyed. Eighteen bridges in Soccsksargen were damaged according to the DPWH initial assessment, with six rendered near-impassable. Bridge failures critically disrupt emergency response, supply chain logistics, and community access in post-earthquake conditions.

| Structure / Location                           | Performance                                 | Failure Mode / Notes                                                                         |
|------------------------------------------------|---------------------------------------------|----------------------------------------------------------------------------------------------|
| St. Elizabeth Hospital, General Santos         | Severe — partial damage; patients evacuated | Critical hospital failure; operations relocated outside — lifeline structure compromise      |
| Notre Dame of Dadiangas Univ. building, GenSan | Severe — partial structural collapse        | Older building vintage; photographic evidence of crumbling confirmed by university president |
| Matanao National High School (condemned bldg.) | Collapse                                    | Pre-condemned structure not demolished; collapse foreseeable. No casualties reported here    |
| Century Pacific Food warehouse wall, GenSan    | Wall collapse — 2 fatalities                | Unreinforced masonry perimeter wall; classic industrial collapse mode                        |
| Barangay Hall, Malalan, Jose Abad Santos       | Collapse                                    | Pre-code small government building; minimal structural supervision                           |
| Bridge, Tapon, Glan, Sarangani                 | Destroyed                                   | Foundation / approach failure; one of 18 bridges damaged in Soccsksargen                     |

|                                                     |                                         |                                                                           |
|-----------------------------------------------------|-----------------------------------------|---------------------------------------------------------------------------|
| SM and Robinsons Malls, General Santos              | Closed for engineering inspection       | Major commercial structures; closed for assessment — no collapse reported |
| National Highway, San Felipe, Tantaran, S. Cotabato | Cracking / damage                       | Road pavement/subgrade failure; ~20 roads rendered impassable             |
| Schools compliant with DepEd standard (post-2010)   | Generally repairable damage             | Protocol-compliant duck-cover-hold evacuations successfully executed      |
| Modern RC frame commercial buildings, GenSan        | Minor to moderate non-structural damage | Structural integrity maintained; non-structural and fitout damage         |

**Table 5.1 — Notable Structural Performance Cases (Sources: PHIVOLCS; Al Jazeera; Philippine Inquirer; GMA News; SunStar Cebu; DepEd DivSur)**

## 6. Engineering Failures: What Went Wrong?

### 6.1 Hospital Seismic Performance — A Systemic Failure

The partial structural damage and functional failure of St. Elizabeth Hospital is the most consequential engineering failure of this event. Hospitals are classified as Importance Category IV structures under IS 1893:2016 and equivalent Philippine NSCP provisions — the highest category, requiring the largest seismic design force multiplier (Importance Factor  $I = 1.5$  under IS 1893). The engineering rationale is self-evident: a hospital must function precisely when an earthquake has disabled it structurally. When a hospital evacuates its patients onto the street during a mass casualty event, the engineering and institutional system has failed at the point it was most needed.

The 2026 event adds to a global record of hospital failures in earthquakes — including the 2010 Haiti, 2015 Nepal, and 2023 Türkiye events — that demonstrates hospital seismic performance is a global engineering crisis, not an isolated event. In India, a significant proportion of hospital infrastructure — particularly government district hospitals and primary health centres in Zone III–V — is pre-code construction without seismic detailing, presenting an identical vulnerability.

### 6.2 The Pre-Condemned School Building — An Institutional Failure

The collapse of the condemned Matanao National High School building represents an institutional and regulatory failure as much as an engineering one. A structure formally condemned after the 2019 Cotabato earthquake series — yet never demolished — remained on an occupied school campus until it collapsed in 2026. This failure pattern — condemned but not removed — is documented in multiple post-earthquake investigations globally. Condemnation without timely demolition or isolation is not risk mitigation; it is risk deferral with added liability.

### 6.3 Unreinforced Masonry Walls in Industrial Structures

The collapse of a warehouse perimeter wall killing two workers is consistent with a failure mode observed in virtually every significant Philippine earthquake. Unreinforced or minimally reinforced masonry perimeter and parapet walls of industrial buildings, warehouses, and commercial structures are

among the most dangerous non-structural elements in seismic zones. They carry little gravity load, receive minimal engineering attention, and are rarely checked for out-of-plane seismic stability. Yet when they fall, they fall onto people.

## 6.4 Infrastructure Vulnerability — Bridges and Roads

Eighteen bridges damaged and 41 roads affected across Soccsksargen within hours of the earthquake reveals the fragility of Mindanao's infrastructure network under seismic loading. Bridges in the Philippines — particularly older structures constructed before modern seismic bridge design provisions — are vulnerable to: (a) approach embankment failure due to liquefaction or lateral spreading; (b) unseating of simply supported spans; (c) pier shear failure; and (d) abutment sliding or rotation. Post-earthquake bridge damage creates a logistics crisis that directly impedes rescue, supply distribution, and medical evacuation.

## 6.5 Tsunami Warning — Engineering Opportunities

The tsunami warning system worked. PHIVOLCS issued warnings promptly. The Pacific Tsunami Warning Center tracked the event. Evacuations were ordered. But the fundamental physical constraint remains: at distances of 20–30 km from a Cotabato Trench rupture, the tsunami arrives within 20–30 minutes of the earthquake. This is the engineering and governance challenge: the warning infrastructure must be so seamlessly integrated into community preparedness that evacuation is reflexive, not deliberative. In this regard, the school teachers who immediately directed duck-cover-hold and campus evacuation — without waiting for official guidance — are the exemplars of the system working as designed.

### ENGINEERING INSIGHT — Hospital Seismic Performance is a Global Crisis

WHO estimates that 1 in 10 hospitals in earthquake-prone countries will be non-functional after a moderate-to-strong earthquake. The 2026 Mindanao event added St. Elizabeth Hospital to this global list. India has approximately 70,000+ government health facilities in Seismic Zones III–V. The proportion with seismic design, peer-reviewed structural drawings, and enforced construction supervision is unknown — but evidence from past events suggests it is very low. Mandating seismic performance of hospitals is not optional; it is a direct function of a government's capacity to respond to the disasters it knows will occur.

# 7. Engineering Success Stories and What They Teach Us

## 7.1 Schools That Used Duck-Cover-Hold

The most documented success story from 8 June 2026 was not architectural — it was behavioural. At Mahayahay Elementary School in Malita, Davao Occidental, teachers calmly directed students to perform the duck-cover-hold protocol during the earthquake. No injuries were reported. This outcome — hundreds of children at their first school day assembly, experiencing the strongest Philippine earthquake since 1990, without a single casualty — is the direct result of trained and practised earthquake response procedures.

This result carries a profound engineering message: the most effective near-term life-safety intervention in Philippine and Indian schools may not be structural retrofitting alone, but the combination of structural adequacy and mandatory, regularly practised emergency response procedures.

## 7.2 SM and Robinsons Malls — Code-Compliant Performance

SM and Robinsons Malls in General Santos City were closed for structural engineering inspections following the earthquake — but they were not reported as collapsed or severely damaged. This reflects the standard practice for major commercial operators in the Philippines: structural engineering inspections before re-opening. The absence of reports of collapse or life-threatening damage at these large, modern commercial structures is consistent with their post-2000 construction and compliance with engineering requirements of major retail developers.

## 7.3 Multi-Agency Response Coordination

The speed and coordination of the multi-agency response — PHIVOLCS issuing the tsunami warning within minutes; AFP deploying HADR teams from multiple brigades; PAF deploying aircraft from Davao; DPWH deploying engineers immediately; President Marcos visiting by 10 June; NDRRMC providing structured situation reports — demonstrated meaningful improvements in Philippine disaster response capability compared to historical benchmarks. This institutional capacity is itself an engineering and governance achievement.

### LESSONS FOR INDIA — Train to Survive, Not Just Design to Survive

The teachers at Mahayahay Elementary School, Davao Occidental did not need a retrofitted building to save their students' lives on 8 June 2026 — they needed training and practice. India's National School Safety Programme (NSSP) provides the framework; implementation at state and district levels remains incomplete. In every Zone III–V school in India, regular earthquake drills, teacher training in duck-cover-hold, and evacuation route planning should be as non-negotiable as the structural design of the building itself. These cost almost nothing. They save lives immediately.

# 8. Comparison with Indian Conditions

## 8.1 Overview: The Cotabato Trench and India's Subduction Zones

The 2026 Mindanao earthquake was a subduction zone event — a mechanism that distinguishes it from the 2025 Cebu strike-slip earthquake. India faces analogous subduction zone threats from: (a) the Andaman–Sumatra subduction system, which generated the catastrophic Mw 9.1 2004 Indian Ocean earthquake and tsunami; (b) the Himalayan inter-plate collision zone capable of Mw 8+ thrust events; and (c) the Makran Subduction Zone off the coast of Balochistan and Gujarat, capable of generating both major earthquakes and tsunamis affecting India's western coast.

| Factor                   | Philippines — Mindanao 2026                                          | India — Comparable Context                                                                                            |
|--------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Event Type               | Subduction zone thrust — Cotabato Trench (Mw 7.8)                    | Himalayan thrust zone; Andaman-Sumatra subduction; Makran subduction                                                  |
| Tsunami Threat           | Confirmed 1–1.4 m waves; 9-province warning                          | Andaman Islands, Tamil Nadu coast (2004 precedent); Gujarat coast (Makran zone)                                       |
| Hospital Failure         | St. Elizabeth Hospital partially damaged — patients evacuated        | Widespread pre-code government hospitals in Zone III–V                                                                |
| School Vulnerability     | 6,000 buildings require assessment; condemned school collapses       | Millions of school buildings; many pre-code; National School Safety Programme under-implemented                       |
| Bridge and Road Failure  | 18 bridges damaged; 41 roads affected; 6 near-impassable             | NH and SH bridges in NE India / Himalayas — largely pre-seismic-code vintage                                          |
| Landslide Hazard         | 6+ landslides; Glan (14 dead); Jose Abad Santos (7 dead, 18 missing) | Uttarakhand, Himachal Pradesh, NE India — extreme landslide susceptibility in seismic terrain                         |
| Industrial Wall Collapse | Warehouse wall kills 2 (Century Pacific Food, GenSan)                | Widespread industrial shed and godown masonry parapet vulnerability in GIDC-type estates                              |
| Pre-condemned Buildings  | Condemned school not demolished — collapses 2026                     | Numerous municipality-tagged 'dangerous' buildings remain occupied across Indian cities                               |
| Early Warning System     | PHIVOLCS warning issued promptly; regional coordination effective    | India's INCOIS tsunami warning system operational since 2007; offshore seismic network requires further densification |

**Table 8.1 — India–Philippines 2026 Seismic Risk Comparison**

## 8.2 Critical Parallels for India

### Hospitals in Seismic Zones

India has an estimated 70,000+ government health facilities, of which a substantial proportion are located in Seismic Zones III–V. A 2015 WHO–MOHFW assessment found that a majority of district hospitals in high-hazard states were constructed before modern seismic codes were adopted. The National Disaster Management Authority (NDMA) has published seismic guidelines for hospitals; mandatory implementation with third-party structural inspection remains the missing link. The St. Elizabeth Hospital failure in General Santos is a direct preview of what awaits these facilities in the next major Indian earthquake.

## School Buildings

India's Sarva Shiksha Abhiyan and subsequent school infrastructure programmes have constructed hundreds of thousands of school buildings across all seismic zones. The quality and seismic detailing of these buildings vary enormously by state, contractor, and year of construction. The fact that 6,000 Philippine school buildings required assessment after a single earthquake event — and that a condemned building was still standing on a school campus — reflects governance challenges identical to those present in India's school infrastructure management.

## Tsunami Preparedness on Indian Coastlines

India's 7,517 km coastline includes significant stretches exposed to tsunami hazard from the Andaman–Sumatra zone (affecting the eastern coast, Andaman Islands) and the Makran zone (affecting Gujarat and Maharashtra western coast). The 2004 Indian Ocean tsunami's 10,749 Indian fatalities established the catastrophic potential. India's Indian National Centre for Ocean Information Services (INCOIS) operates the Indian Tsunami Early Warning System; effectiveness depends critically on community-level preparedness and the 20–30 minute warning windows available at proximate coastlines.

### LESSONS FOR INDIA — The Condemned Building Problem

India's municipal corporations and gram panchayats identify structurally unsafe buildings ('dangerous buildings') regularly, yet removal or occupant relocation is often delayed by years or decades due to legal, financial, and political constraints. The collapse of the pre-condemned Matanao National High School building is the precise consequence of this deferral. India needs a legally enforceable 'condemned = vacated within 90 days' framework with government relocation assistance — not merely a paper classification system.

## 9. Key Lessons for Structural Engineers

| Category            | Lesson                                                                                                                                                | Reference / Standard                                       |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| Design — Hospitals  | Design all healthcare facilities to Importance Category IV ( $I = 1.5$ ); use PBEE to demonstrate post-earthquake functionality, not just life safety | IS 1893:2016 Table 6; FEMA P-58; WHO Safe Hospitals        |
| Design — Bridges    | Design bridges with collapse-prevention performance objective for DBE + MCE; provide seismic isolation for critical bridges in Zone IV/V              | IRC:6-2017 Cl. 222; AASHTO LRFD Seismic; Eurocode 8 Part 2 |
| Design — Industrial | Design all masonry parapet and perimeter walls in industrial buildings for out-of-plane seismic forces; provide positive anchorage to framing         | IS 1893:2016 Cl. 10; IS 13920:2016 Cl. 9.3                 |

|                    |                                                                                                                                                                                    |                                                           |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Design — Tsunami   | For structures within Coastal Zone Management Act zones in tsunami-prone areas: design for combined seismic + tsunami force; provide elevated refuge or vertical evacuation routes | FEMA P-646; ASCE 7-22 Ch. 6; IS 4651 (Coastal Structures) |
| Detailing          | Seismic ties, 135° hooks, and confinement reinforcement are non-negotiable in all structures in Zone III–V — regardless of structure type or scale                                 | IS 13920:2016; ACI 318-19 Sec. 18                         |
| Infrastructure     | Assess all bridges in Seismic Zone III–V using IS 1893:2016 and IRC:6-2017; replace simply supported spans with continuous framing or provide hold-down devices                    | IS 1893:2016 Part 3 (Bridges); IRC:6; AASHTO              |
| Institutional      | Enforce demolition of formally condemned structures within a defined statutory period; condemned = vacated, not merely labelled                                                    | National Building Code India 2016; Municipal Acts         |
| Preparedness       | Make earthquake drills mandatory and quarterly in all schools, hospitals, and government buildings in Zone III–V — equal in importance to structural design                        | NDMA 2007 Guidelines; NSSP; NBC India 2016                |
| Site Investigation | For coastal structures: assess tsunami run-up elevation and assess combined seismic + tsunami load; include in geotechnical report                                                 | IS 1893:2016 Annex F; FEMA P-646                          |
| Supervision        | Hospital and school construction must include mandatory structural engineer site inspections; no occupancy certificate without engineer certification                              | NBC India 2016 Part 6; IS 456:2000                        |

**Table 9.1 — Key Engineering Lessons with Applicable Standards**

## 10. Implications for Architects and Developers

### 10.1 Coastal Development and Tsunami-Resilient Design

The 2026 Mindanao earthquake adds urgent relevance to the question of coastal development standards in tsunami-prone zones. The Philippines — and India — have both experienced major tsunamis within living memory (2004 Indian Ocean; 1976 Moro Gulf). Yet coastal development continues in high-hazard zones without systematically incorporating tsunami-resilient design principles.

Tsunami-resilient design principles for coastal structures include:

- Elevate occupied floor levels above the design tsunami run-up elevation (determined by site-specific hazard analysis)



- Use open-ground-floor or pilotis construction to allow tsunami flow-through beneath occupied levels — paradoxically, the structural arrangement most dangerous for earthquakes (soft storey) provides the least obstruction to tsunami inundation when correctly designed
- Use reinforced concrete or structural steel framing in preference to masonry for the lowest levels of tsunami-exposed structures
- Provide vertical evacuation structures (VES) — reinforced concrete towers capable of resisting both seismic and tsunami hydrodynamic forces — in communities where horizontal evacuation to high ground is not achievable within available warning time
- Remove parapet walls, signage, and unreinforced facade elements that become deadly debris in tsunami waves

## 10.2 The Multi-Hazard Design Imperative

The 2026 Mindanao earthquake was not a single-hazard event. It was simultaneously: an earthquake causing structural damage; a tsunami generator causing coastal inundation; a landslide trigger in mountainous terrain; and a lifeline infrastructure disruptor. Engineers and developers in comparable settings — India's Andaman Islands, coastal Tamil Nadu, coastal Gujarat, and the Himalayan belt — must design for multiple simultaneous hazards, not sequential single-hazard evaluation. Multi-hazard probabilistic risk assessment is available and should be required for all critical and large-scale development in high-hazard zones.

### FIGURE SUGGESTION — Figure 10.1

Schematic cross-section of tsunami-resilient coastal building showing: design tsunami run-up elevation, elevated occupied floor, open ground floor allowing flow-through, foundation system anchored to bedrock or piles, vertical evacuation staircase, and structural framing resisting combined seismic and hydrodynamic forces. Source: FEMA P-646 / ASCE 7-22 Ch. 6.

## 11. Role of Modern Technologies

### 11.1 Tsunami Early Warning Systems — Technology and Limits

The PHIVOLCS tsunami warning system demonstrated effectiveness in the 2026 event — issuing warnings promptly with a regional coordination network encompassing BMKG Indonesia, PTWC USA, and multiple Pacific rim agencies. However, technology cannot overcome physics: a Mw 7.8 Cotabato Trench earthquake generates a tsunami that reaches the Sarangani coastline within approximately 20–30 minutes. For communities within this window, the warning system's value is entirely dependent on pre-trained, pre-practised community response — not on the warning technology itself.

### 11.2 Performance-Based Design for Multi-Hazard Scenarios

Traditional prescriptive seismic design codes address earthquake loads in isolation. Modern performance-based frameworks — FEMA P-646 for tsunami loads, ASCE 7-22 Chapter 6 for tsunami design of buildings, and combined seismic-tsunami probabilistic analysis methods — allow engineers to

quantify structural risk under combined loading scenarios that closely reflect real-world conditions like those experienced in the 2026 Mindanao event.

### 11.3 AI-Assisted Damage Assessment and UAV Surveys

Following the 2026 event, drone deployment by the PAF Tactical Operations Group 11 for rapid aerial damage assessment represented a significant operational improvement over ground-only surveys. Integration of AI-based damage classification algorithms — trained on multi-event global imagery datasets — with drone-captured imagery can generate preliminary damage maps within hours of an event, enabling prioritisation of engineering inspection resources. This technology is rapidly maturing and should be built into national rapid assessment protocols.

### 11.4 Structural Health Monitoring for Lifeline Structures

The partial failure of St. Elizabeth Hospital reinforces the case for permanent structural health monitoring in all Category IV (Importance Factor 1.5) structures in Seismic Zones III–V. A network of accelerometers and displacement sensors installed in a hospital building can provide automated damage state assessment within minutes of a seismic event, enabling clinical staff to make informed decisions about evacuation versus continued operation based on real structural response data rather than visual inspection or precautionary assumptions.

## 12. Recommendations for Indian Engineering Practice

### 12.1 Design Office

- For all hospitals and health facilities in Zone III–V: use IS 1893:2016 Importance Factor  $I = 1.5$ ; conduct PBEE assessment demonstrating post-earthquake operational continuity, not merely structural survival
- For all bridges in Zone III–V: perform IS 1893 Part 3 (Bridges) and IRC:6-2017 seismic assessment; prioritise retrofit of pre-code simply-supported bridges on soft or liquefiable foundations
- For coastal structures in tsunami-hazard zones: apply FEMA P-646 / ASCE 7-22 Ch. 6 tsunami design provisions; determine site-specific design tsunami run-up elevation using INCOIS hazard data
- For all industrial structures with masonry parapet or perimeter walls: design explicitly for out-of-plane seismic forces per IS 1893:2016 Cl. 10; provide positive mechanical anchorage to framing
- For all multi-storey buildings in Zone IV/V: verify no pre-condemned or structurally unsafe elements exist on the campus before occupancy renewal

### 12.2 Site Execution

- Hospitals and schools in Zone III–V: mandate structural engineer certification at foundation, rebar, concrete, and facade stages — no occupancy certificate without signed engineer inspection records

- Industrial buildings: inspect all masonry parapet and boundary wall construction for absence of horizontal bond beam reinforcement — install if absent
- Coastal structures: verify finished floor elevations against design tsunami run-up height before occupancy; document in structural completion certificate
- All RC structures: enforce 135° seismic hooks on all column and shear wall ties — photograph before concrete pour

### 12.3 Audit and Rehabilitation

- Conduct immediate seismic audit of all government hospitals and health centres in Zone III–V using a standardised rapid visual screening protocol — this is a national life-safety emergency, not a future planning exercise
- Formally demolish, not merely condemn, all structurally unsafe buildings on occupied public premises (schools, hospitals, government offices) within a legislated time period
- Assess all bridges on National and State Highways in Zone III–V for seismic vulnerability; prioritise retrofit of bridges on river crossings where liquefaction or lateral spreading is possible
- Assess industrial estates and GIDC-type zones for unreinforced masonry parapet and perimeter wall vulnerability; issue advisory notices to industrial associations in Zone IV/V areas

## 13. Expert Commentary

The following statements are drawn from verified published sources relating to the 2026 Mindanao earthquake:

*"This is the strongest earthquake we've experienced."*

— Local police chief, General Santos City, as quoted in CNN coverage of 08 June 2026.

*"The national government is moving and we will not leave Mindanao behind."*

— President Ferdinand Marcos Jr., 08 June 2026, directing all government agencies to respond immediately.

*"Based on the location of the earthquake, focal mechanism, and aftershock distribution, the earthquake may be attributed to subduction along the Cotabato Trench."*

— PHIVOLCS/DOST Official Primer, 08 June 2026, 11:00 AM PST.

The Miyamoto International rapid assessment team, deploying within hours of the event, reported: 'The mainshock was followed by a significant aftershock sequence, including events of M6.5, M6.0, and M6.0, along with multiple magnitude-5 events clustered near Sarangani' — confirming the extended aftershock hazard that complicates both rescue operations and structural re-assessment.

These verified expert statements — from the national president to the seismology agency to international earthquake engineers — paint a consistent picture: this event was seismically predictable (Cotabato Trench is a known major hazard), the response was mobilised, but the structural and institutional failures that produced casualties were preventable with adequate enforcement of existing engineering standards.

### **LESSONS FOR INDIA — The Cotabato Trench Lesson for the Makran Zone**

The Makran Subduction Zone off Pakistan-Iran generates thrust earthquakes and tsunamis directly analogous to the 2026 Mindanao event. The 1945 Makran earthquake (Mw 8.1) killed approximately 4,000 people and generated a tsunami that struck the Gujarat and Sind coasts. Gujarat's coastline — including the ports of Kandla, Mundra, and Hazira — and Mumbai's western-facing coast are within the potential tsunami inundation zone of a major Makran event. India's INCOIS warning system covers this threat; community preparedness, coastal development controls, and tsunami-resilient design of coastal infrastructure are the engineering responses that must match the warning technology.

## **14. The Future of Seismic and Tsunami Resilience**

### **14.1 Recommendations for Government**

- Make structural assessment and certification of all government hospitals in Seismic Zones III–V a national emergency programme with a five-year completion deadline
- Enact and enforce a statutory time limit for demolition of formally condemned structures on occupied public premises
- Expand tsunami vertical evacuation infrastructure in communities within 30-minute tsunami inundation windows on the Andaman Islands, Tamil Nadu coast, and Gujarat coast
- Mandate seismic assessment of all bridges on National Highway and State Highway networks in Zone III–V; publish results and retrofit funding requirements
- Integrate combined seismic-tsunami hazard mapping into coastal zone management plans; publish INCOIS hazard maps at 1:50,000 scale for all coastal districts
- Fund community earthquake and tsunami drills in all schools and hospitals in Zone III–V — not as one-time events but as mandatory quarterly exercises

### **14.2 Recommendations for Industry**

- Major retail, commercial, and industrial developers in Zone III–V: implement mandatory post-earthquake structural assessment protocols before re-opening; publish protocols publicly
- Insurance industry: develop seismic performance-linked premium structures for industrial facilities to incentivise unreinforced masonry wall upgrades
- Coastal resort and hotel developers: commission tsunami risk assessments from qualified coastal engineers before designing or expanding beachfront facilities in tsunami-hazard zones

- Adopt FEMA P-646 tsunami design provisions for new coastal construction in areas with documented tsunami history

### 14.3 Recommendations for Academia

- Include subduction zone earthquake and tsunami engineering as a required module in undergraduate structural and geotechnical engineering curricula
- Conduct joint post-earthquake investigation teams with ASEP, EERI, and SEAOC for all future major Philippine earthquake events — India's IITs and NIT earthquake engineering groups should formally participate
- Develop India-specific tsunami fragility functions for coastal building types — currently absent from the literature
- Research performance of Indian coastal industrial infrastructure under combined seismic + tsunami loading — a major knowledge gap

## 15. Conclusion

The 08 June 2026 Mindanao earthquake — Mw 7.8, the most powerful to strike the Philippines in 36 years — killed 47 people, injured 688, left 33 missing, damaged or destroyed more than 12,600 houses, and disabled critical infrastructure across southern Mindanao. It struck at 7:37 in the morning on the first day of the school year, when children and teachers across Mindanao were conducting flag ceremonies. It also generated a tsunami, triggered at least six landslides, and damaged a functioning hospital and two universities.

The engineering record from this event is, once again, consistent with every major earthquake before it. Structures that failed were old, pre-code, or structurally condemned but undemolished. Structures that performed well were modern, code-compliant, and well-maintained. Schools where teachers practised duck-cover-hold had no casualties. Schools where condemned buildings remained standing had collapses. Hospitals that were seismically vulnerable became crisis centres themselves during the disaster response.

These patterns are not unique to the Philippines. They are the universal patterns of earthquake vulnerability — and they are present, at scale, in India's seismic zones. India's next major earthquake will ask the same questions of its engineers, developers, regulators, and institutions that the 2026 Mindanao event asked of the Philippines. The answers depend on decisions being made today: in design offices, on construction sites, in municipal approval offices, and in engineering education programmes.

The lesson is not that earthquakes are unpredictable or unstoppable. It is that their consequences are largely engineered — by the quality of what we build, the rigour with which we maintain it, and the seriousness with which we prepare for what we know is coming.

#### THE ENGINEER'S COMMITMENT — FormX Consultants

At FormX Consultants, we approach every structural design project — from foundations to facades — with the knowledge that the structures we engineer must perform under the

worst conditions our region's seismicity can impose. Every reinforcement detail, every site inspection, every geotechnical recommendation is an act of professional responsibility. The 2026 Mindanao earthquake is the latest reminder that engineering integrity is not a quality preference — it is a life-safety obligation.

## STRUCTURAL ENGINEER'S EARTHQUAKE & TSUNAMI PREPAREDNESS CHECKLIST

Based on Lessons from the 2026 Mindanao (Mw 7.8) and 2025 Cebu (Mw 6.9) Philippines Earthquakes  
Applicable to Indian Practice — IS 1893:2016 / IS 13920:2016 / IS 15988:2013 / IRC:6-2017 / FEMA P-646

Prepared by **Hiren J Shah, Structural Engineer** | **FormX Consultants — Design | Engineering**

### A. DESIGN PHASE — ALL STRUCTURE TYPES

- ☐ Confirm Seismic Zone and site class per IS 1893:2016; conduct site-specific PSHA for Zone III–V projects in urban or critical use categories
- ☐ Classify all structures by Importance Category (IS 1893 Table 6); apply corresponding Importance Factor ( $I = 1.0$  to  $1.5$ )
- ☐ Hospitals, Emergency Centres, Schools:  $I = 1.5$  minimum; evaluate post-earthquake operational continuity, not just life safety
- ☐ Check structural regularity in plan and elevation per IS 1893 Tables 4 and 5; apply penalties and dynamic analysis for irregular buildings
- ☐ Identify all short-column conditions from partial-height masonry infill; design for short clear-height shear demand explicitly
- ☐ Design all masonry parapet, perimeter, and infill walls for out-of-plane seismic forces per IS 1893:2016 Cl. 10
- ☐ For coastal structures: determine design tsunami run-up elevation from INCOIS hazard data; apply FEMA P-646 / ASCE 7-22 Ch. 6 tsunami design provisions
- ☐ For bridges: apply IS 1893 Part 3 (Bridges) and IRC:6-2017; evaluate liquefaction and lateral spreading risk at all river crossings in Zone III–V
- ☐ For coastal industrial / warehouse structures: design for combined seismic + hydrodynamic loading if within tsunami inundation zone

### B. DETAILING PHASE (per IS 13920:2016)

- ☐ Columns:  $135^\circ$  seismic hooks on ALL ties — maximum spacing =  $\min(d/4, 6db, 100 \text{ mm})$  in plastic hinge zones — non-negotiable
- ☐ Beams: stirrups at  $\leq \min(d/4, 8db, 150 \text{ mm})$  in plastic hinge zones;  $135^\circ$  hooks; minimum 2 bars top and bottom continuous
- ☐ Beam-column joints: full shear reinforcement through joint panel zone per IS 13920:2016 Cl. 11

- ☐ Shear walls: boundary elements with closely spaced confinement ties where extreme fibre compression exceeds  $0.2 f_{ck}$
- ☐ Masonry infill: provide isolation gap  $\geq 25$  mm OR reinforced infill per IS 13920 to prevent unintended frame-infill interaction
- ☐ Industrial masonry walls: provide vertical and horizontal bond beam reinforcement; anchor to RC framing at all contact points
- ☐ Coastal structures in tsunami zone: design ground-level columns / piles for tsunami hydrodynamic and debris impact forces per FEMA P-646 Section 6

### C. CONSTRUCTION SUPERVISION

- ☐ Hospitals and schools: minimum two formal structural engineer site inspections per floor cycle; certify compliance in writing
- ☐ Verify rebar, spacing, hook formation, and cover before every pour; photograph and archive records
- ☐ Test concrete: minimum 3 cubes per  $5 \text{ m}^3$ ; reject batches below  $0.75 f_{ck}$  at 7 days; maintain test register
- ☐ Parapet and perimeter wall construction: inspect for bond beam reinforcement and anchorage continuity before plastering
- ☐ Coastal structures: verify finished floor elevation against design tsunami run-up height before occupancy
- ☐ No occupancy certificate for hospitals, schools, or emergency facilities in Zone III–V without structural engineer completion certificate

### D. EXISTING BUILDING ASSESSMENT AND RETROFIT

- ☐ Hospitals and health centres in Zone III–V: conduct IS 15988:2013 / FEMA 154 Rapid Visual Screening immediately; escalate all Level 2 structures to detailed assessment
- ☐ School buildings: apply NDMA / NSSP rapid visual screening to all pre-2000 buildings; assess all condemned or 'dangerous' tagged structures for immediate vacation and demolition
- ☐ Bridges: conduct IS 1893 Part 3 seismic vulnerability assessment for all pre-2000 river crossing bridges in Zone III–V
- ☐ Industrial facilities: inspect all masonry parapet and perimeter walls for absence of seismic anchorage; retrofit with anchor straps or RC bond beam if absent
- ☐ Formally condemned structures: initiate vacation, legal closure, and demolition proceedings — do not allow continued occupation or campus co-existence with occupied structures

### E. TSUNAMI AND MULTI-HAZARD PREPAREDNESS

- ☐ For structures within INCOIS tsunami inundation zones: obtain and display design tsunami run-up elevation on building drawings and occupancy certificates



- ☐ For coastal public buildings and critical facilities: install tsunami evacuation wayfinding signage per NDMA guidelines
- ☐ For schools, hospitals, and government buildings within 30-minute tsunami travel time of a subduction zone: conduct tsunami evacuation drills biannually
- ☐ Ensure vertical evacuation routes (staircase to roof or upper floors above tsunami run-up) are structurally capable of resisting combined seismic and tsunami loading
- ☐ Coordinate with INCOIS / NDMA to ensure coastal development projects incorporate tsunami risk assessment before design commences

## F. POST-EARTHQUAKE RESPONSE

- ☐ Conduct Rapid Visual Assessment of all structures after any earthquake with felt intensity  $\geq$  IV using ATC-20 / IS 13935 protocol
- ☐ Apply Green / Yellow / Red placard system; do not authorise re-occupancy of Yellow or Red tagged structures without licensed structural engineer certification
- ☐ For hospitals: post-earthquake structural assessment must be completed before resuming full operations — use SHM data if available to accelerate decision
- ☐ File post-earthquake damage report with state SDMA within 30 days; retain records for national damage database
- ☐ For tsunami events: assess coastal structures for combined wave and seismic damage before re-occupancy; engage coastal engineer for structures within inundation zone

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